

Coursework Specifications.

Comp1939 – Ethics in Data Science.
2025-26.

Module Leader: Dr Stef Garasto and Dr Makuochi Nkwo.

Deadline Date: 31/03/2026, 17:00 UK time.

Contribution: 100% of course.

Pass mark: 50%+.

Student Workload: This coursework is expected to take an average student who is up to date with tutorial work approximately 50 hours to complete.

Marking: Feedback and grades are normally released within 17 working days (excluding bank holidays, weekends and university closure days) of the coursework deadline. Your coursework will be marked in a pseudonymised format: you will need to include the group number, but do not include any indication of your individual identity in your submission.

Learning Outcomes:

1. Critically synthesise and analyse a variety of critical, societal and ethical perspectives on the impact and drivers of data-driven technologies.
2. Critically evaluate the ethical risks associated with specific Data Science use cases and formulate potential mitigation strategies.
3. Apply knowledge of data-related legislation and governance practices to specific Data Science applications and evaluate their suitability for those applications.

Plagiarism and AI declaration.

Plagiarism is presenting somebody else's work as your own. All sources used need to be referenced correctly. Suspected plagiarism will be investigated and if found to have occurred will be dealt with according to the procedures set down by the University. Please see your student handbook for further details of what is / isn't plagiarism.

Guidance to the AI usage and the university policy and student declaration: [guidance to AI usage](#); [declaration of AI use](#). We recommend giving lots of thought to whether AI technologies are suitable for use with this assignment, since results can easily be generic and superficial, or even wrong. Using AI is not prohibited if it doesn't replace your learning, but we recommend always critically reflecting on the output. Indeed, if you do use generative AI, you need to: a) declare, reference and quote it appropriately (otherwise it's an academic offence); b) reflect on which (and whose!) ethical perspective(s) might be influencing the AI's suggestions.

Coursework Submission Requirements.

For your coursework submission to be admissible, an electronic copy of your work for the coursework must be fully uploaded on the **Deadline Date of 31st March 2026 (17:00 UK time)** using the relevant link on COMP1939 Moodle page.

Your coursework consists of **two deliverables**. Both files are to be uploaded in the same (individual) submission link. **A wrongly named file may incur a mark penalty**. The two files are:

- **A single PDF file named 'report_comp1939_groupX'**, which is the **individual report** with the case study analysis of 2500 words (+10%). If the submitted work exceeds the limit by more than 10%, marks will be deducted. In the file name, replace X with your group number. Do not put your name or student ID in the report.
- **A single PDF file named 'poster_comp1939_groupX'**, containing the group poster (exported from the template). In the file name, replace X with your group number. The word count for the poster is 800 words (+10%). **Each member of the group will have to submit the same poster: this is by default so it will not be considered plagiarism.**

Before that, you need to take some other actions as outlined here:

1. **Form your group (4 people) by 06/02/2026**. Add the group details to the spreadsheet linked from Moodle. Those not in a group by that date will be randomly allocated to one.
2. **Upload a description of your chosen case study, by 20/02/2026** using the relevant Moodle submission link (**one submission per group**, any member of the group can upload it). A short description (150 - 300 words) will suffice. Please describe your group's case study briefly, one or two examples of potential ethical risks and/or benefits and, if you have one, link to an external resource that may help support your analysis. We will give **formative** feedback on all submissions uploaded by the stated date.

Please note:

- Any text in the document must not be an image (i.e. must not be scanned).
- There are limits on the file size (see the relevant course Moodle page).
- Make sure that any files you upload are virus-free and not protected by password or corrupted otherwise they will be treated as null submissions.
- You must NOT submit a paper copy of this coursework.
- Lecturers and tutors can **only** accept coursework submitted via the appropriate Moodle link. No other way of submitting the coursework will be considered.

The University website has details of the current Coursework Regulations, including details of penalties for late submission, procedures for Extenuating Circumstances, and penalties for Assessment Offences. See [University policies](#).

Detailed Coursework Specification.

You have just joined the ethics advisory board of a non-departmental public body focused on the use of data-driven services within government. The committee's role is to advise on the ethical and societal implications of new and existing data projects undertaken within the public body, to understand what changes need to be made to increase the projects' societal benefits, or even, on occasions, if a project should go ahead at all.

Since the committee is newly formed, the group decides to start by **developing an analysis portfolio of past data-driven projects or specific technologies used as case studies**. So, the group decides that 4 people will analyse the same case study and compare findings.

It is decided that there is a broad range of what can be used as a case study, and that this could be a project¹, a product or a technology. The main constraint is that **it needs to be specific and a case study of a project that is “data-driven”**. That is, the project you choose to analyse needs to be primarily focused on working with data, potentially at large scale, and it needs to involve some elements of statistical analysis, and/or the generation of a score, a ranking, or a prediction from the data, typically using a math-based algorithm. If looking at a data-driven technology, the analysis should be of this technology within a **specific context** (for example, not just “chatbots”, but “chatbots for customer service within private companies”). However, each group member can choose a different context. A case study can be from anywhere in the world and from any organisation, team, company, governmental body, etc. Of particular interest are case studies characterised as being “for the social good”, with documented harms or controversies, or government related. To make the starting point easier, the committee collects and shares a list of [potential case studies](#) (accessible at the link provided).

For each project, you make it your task to analyse how “good” and “harms” are conceptualised within the project, as viewed through the lens of existing “ethical toolkits” and specific ethical and/or sociological theories. The analysis should also touch upon who was impacted by the benefits or harms, how relevant communities have been or could have been involved, and relevant legal and governance concerns. It should also present practical recommendations. Finally, you want to understand how comparing multiple perspectives for the same case study can enrich the ethical analysis.

So, your role is to **work as a group to identify and analyse a case study falling within the scope outlined above**. You will produce a **written report individually**, and a **group poster**. The non-departmental public body and its stakeholders are a mix of people with various fields of expertise, so we recommend using the “jargon” of the topic of the project as little as possible and explaining relevant terms and concepts when and if they need to be introduced.

Each person in the group will **conduct the analysis from their own perspective and/or using different tools: this will form the basis of your individual report**. You will then work together to **compare and integrate your analyses into one group poster**. You can exchange ideas and discuss in your group, but it is recommended to do this while taking no or minimal notes, so that the discussion will most likely be filtered through your own perspective when writing.

Individual report content.

The content of the report needs to include the elements outlined below, although you can adapt

¹ We will refer to it only as “project” in the rest of the specifications for brevity, unless needed otherwise.

the exact structure to your needs. Please pay particular attention to point 6, since it is a relatively unusual one and will have a large influence on your mark.

1. **Summary and framing.** First, summarise the project by giving details on: project name, project objectives and its main computational outcome (e.g. its score or prediction), when and where it took place, teams/funders involved, datasets used, any other (small) details relevant for the following discussion. When something is not (fully) known, you can say so or add the partial information that it is known. Then, briefly frame and summarise the line of argument forming the rest of the report.
2. **Conceptualisation of good/harms.** Critically discuss how “social impact” is intended within this project, why you think this project contributes to “social good” (or to what extent it does), while discussing the way it possibly harms people. When discussing these aspects, pay particular attention to the notion of “**whose social good**” and “**who is/may be harmed**”. Who are the supposed users/beneficiaries of the project, or who is meant to be generally impacted by the project, according to the developers/researchers/funders, and how? Are there more groups impacted than what envisioned by the project’s objectives? Are the benefits/harms as envisioned (or not) and how were those benefits/harmed proved?
 - a. Please use **one (ethical) risk assessment toolkit and one or more ethical or sociological frameworks** from class to structure your discussion. For example, for the latter you may use ethical theories and/or considerations of power dynamics and/or others. For the selected toolkit, you are free to only use part of it, as long as your selection is reasonable and justified.
 - b. You can use personal reflections based on your own experience alongside arguments sourced from and/or backed up by relevant literature. Please, make it clear when you are using one or the other, but multiple forms of knowledge can be acceptable, if appropriately contextualised.
 - c. You are expected to both refer specifically to contents taught in class **and** use some resources from outside the teaching material.
3. **Communities engagement.** Has the project incorporated participatory approaches? If yes/no, critically discuss the project’s use (or lack thereof) of participatory approaches, including with reference to Arnstein’s ladder of participation and by considering what the benefits and limitations of such engagement would be. Discuss which specific communities the project should engage (or has engaged or has failed to engage) and reflect on how it should do so, as well as issues of potential power imbalances.
4. **Data governance and legal concerns.** What are the data governance principles around which the project is structured? If known, critically discuss the appropriateness of the data governance principles. If not known, outline some data governance principles you would apply to the project and critically discuss why. Provide two specific questions about the project you would like to ask a lawyer, with a commentary on why you prioritised these questions.
5. **Recommendations.** Provide two **practical and specific recommendations** for how to

improve the project from an ethical perspective, based on your discussion above. These are aimed at data scientists that might be working on this project but should not be “technical” (in the sense of being related to the computing/mathematical aspects of the project).

6. **Personal reflection and your chosen weights for the criteria.** There are two elements to this reflection, both of which mandatory:
 - a. Provide a **weight for criteria A, B, C, D and E** in the rubric below that you would like me to apply for your submission. **You can choose any weight you like, provided these are within the range stated in the marking rubric below and they sum up to 100%.** We will use your chosen weights when marking your submission. A weight outside the range will be approximated to the closest range boundary. Weights will be normalised so that they sum up to 100%. If you do not provide your own weights, default ones will be used, as described in the rubric below. If you would have received a higher mark had the default weights been used, that higher mark will be awarded.
 - b. Give a **brief justification** of your choices: why/how did you prioritise certain criteria over others? Also provide your reflections on this initiative with respect to **the broader topic of participatory approaches**. Example questions you could consider are: could this process be said to be participatory? To what extent? Did it change your approach to the assignment or how you felt about it?
7. **References.** Cite references and sources used at the end. These can be academic papers, blogs, and more. Remember to give credit if someone else’s work has helped you complete your coursework. References are excluded from the word count.

Group poster content.

Create a group poster using the [template provided on Moodle](#) where you a) summarise the case study; b) give a summary of each individual analysis of the case study (highly recommended to focus on the most important highlights); c) compare and contrast each other’s perspectives: what did you agree/disagree on? Why? What can we learn or conclude about the case study from this comparison? In addition to being submitted on Moodle, posters can also be printed and exhibited during the class showcase in week 12. The idea is to have a mini conference where you can present your work to the tutors for some early feedback and exchange ideas with your classmates. For more details, please refer to the instructions on Moodle (under “Assessment and coursework submission guidance”).

Word count.

The word count for each **individual report is 2500+10%**. If the submitted work exceeds the limit by more than 10%, marks may be deducted. There is no lower word limit, but your report

still needs to meet all the criteria outlined in the marking rubric below.

The report will likely include the sections below, though you are free to deviate from this structure as long as all the necessary elements are covered:

1. Project summary and framing.
2. Critical discussion on “good” and “harms”.
3. Communities engagement plan (can also be part of the ‘Critical discussion’ section, potentially as a sub-section).
4. Data governance and legal concerns (can also be part of the ‘Critical discussion’ section, potentially as a sub-section).
5. Recommendations – recommended 100 to 250 words.
6. Individual reflection – recommended 150 to 300 words

The group poster should use the [template provided](#), with a word count of **800+10% words in total per group**.

Checklist.

This checklist is to help you follow whether your report meets the overall requirements outlined in the “Individual report content” section. The parenthetical comments at the end of the points here (e.g. Criterion C) refer to the category in the marking rubric by which that aspect of the report will be judged. The marking rubric used to mark your work is at the end of this document (from page 8).

You should ensure you have:

1. Identified and described a suitable project in a clear manner. **(Criterion A)**.
 - a. Talk to us if you are really struggling with this. If you want to, you could use a data project that you came up with but does not (yet) exist.
2. Critically discussed the benefits, harms and risks of the project, from multiple perspectives, stakeholders and theories. **(Criterion C)**.
 - a. The goal is not to arrive at a definitive moral judgement on the project (at times this is possible, other times reality is more nuanced). Instead, one should think through the assumptions, values and goals embedded within the project (implicitly or explicitly) and how they can lead to concrete harms and benefits, and for whom.
3. Referred to all of the 4 required “lenses” of analysis, which are: **notion of good and harms using relevant ethical and/or sociological theories; ethical toolkit; communities engagement; data governance and legal concerns**. **(Criterion A)**.
 - a. How much space you devote to each of them is up to you. Depending on the

focus of your report, you can engage more with one/some of them or treat each equally.

4. Referenced topics from class correctly, critically and in a way that is tailored to your case study. **(Criterion B, C).**
5. Done at least some independent research beyond the material presented in class. The independent research is incorporated coherently and the external sources are referenced properly. **(Criterion B).**
6. Engaged critically with sources/arguments included. How do you know something you have written is true? Would other people disagree? Does it depend on assumptions that we take for granted but might not be true? **(Criterion C).**
7. Referenced all your sources and, if using generative AI tools, discussed how their use might have influenced the content of your work. **(Criterion B).**
8. Structured your work in a logical and coherent manner. How you structure the sections within your report is up to you. **(Criterion A).**
9. Wrote your work in a clear and succinct manner. **(Criterion A).**
10. Focused as much as possible on “concrete” issues, rather than “general/abstract” ones (although there can be a role for the latter too). **(Criterion D).**
 - a. For example, this is the difference between saying “All relevant data needs to be stored securely in accordance with the law” and “This project collects data about people’s gender, but does not use such data at any point. So, according to a data minimisation principle, gender data should not be collected at all”.
11. Checked for typos, grammatical mistakes, etc. **(Criterion A).**
12. Included two suggestions for improvement that are as practical and concrete as possible. These need to follow logically from the arguments presented in your report. **(Criterion D, A).**
 - a. Note that the notion of improvement may include stopping or drastically altering the project, as appropriate.
13. Included a thoughtful self-reflection that is specific to you/your group and your work, giving the weights for each criterion and relevant commentary. **(Criterion F).**
14. Completed a clear and succinct group poster where you summarise the case studies, the main highlights from each of your perspectives, and how they compare to and integrate with each other. **(Criterion E).**

You can pass the course without meeting all the elements of this checklist, or by only partially meeting all/some of them, but the more of them you successfully meet, the better your mark. We have also tried to map the checklist’s elements to the assessment criteria to better show how each of them might influence your mark.

Rubric (assessment criteria).

Please see the table below for more details about how marks will be allocated. Each criterion is scored between 0% and 100%. The final mark is a **weighted average of these scores, rounded to the nearest integer**. You decide your exact weight for criteria A to E (as long as each is within the range and all weights for criteria A to F sum up to 100%). Please provide your weights in the reflection section.

	0-29% Fail	30-49% Fail	50-59% Good	60-69% Very Good	70-79% Excellent	80-100% Exceptional
Criterion A (default: 20%, range: 10% to 30%). Addresses the assigned topic appropriately, with a clear focus, appropriate language and logical structure. Refer to the checklist for certain details, for example the “lenses” are listed under bullet point 3 of the checklist.	Wrong or not suitable topic addressed. The report is largely unintelligible, lacks coherent structure and focus, is overly verbose with too many typos. None of the required analysis “lenses” are included and recommendations are absent or completely disconnected from the rest of the report.	The topic addressed is only partially compatible with specifications. The report is mostly unintelligible, with little coherence and focus, is very verbose with many typos. Only few of the required analysis “lenses” are appropriately included and recommendations are mostly disconnected from the rest of the report.	A suitable topic is addressed. The report is clear enough, with some loose structure and focus, but may still be hard to read. All but very few of the required analysis “lenses” are appropriately included. Recommendations are loosely connected.	A suitable topic is addressed. All but one of the analysis “lenses” are appropriately included. The report is clear. There is a structure and focus, which may still be loose and with hard-to-read parts for a minority of the report. Recommendations are sufficiently connected.	A suitable topic is addressed. All analysis “lenses” are appropriately included. The amount of space given to each of them is suitably justified within the context of the report. The report is very clear, with an explicit focus and a structure that develops the argument logically for most of the report. There may be some minor mistakes. Recommendations are well connected.	A suitable topic is addressed. All analysis “lenses” are appropriately included. The amount of space given to each of them is perfectly justified within the context of the report. The report is perfectly clear, with a distinct focus. It has a clear structure, with the argument developed logically throughout all the report. Recommendations are perfectly connected.

	0-29% Fail	30-49% Fail	50-59% Good	60-69% Very Good	70-79% Excellent	80-100% Exceptional
Criterion B (default: 20%; range: 15% to 30%). Demonstrates the knowledge of relevant ethical topics and issues for the case study, and references sources appropriately.	No relevant material is used from the module or from outside the module. There is little to no content that demonstrates an understanding of relevant topics.	Uses some of the sources and materials from the module, but applies them incorrectly, and/or misunderstands the theories that are being used.	Draws on some relevant theoretical and practical tools related to ethics in data science (possibly including own experience). These tools are mostly used correctly, but there can be some mistakes. There may be one source not from teaching material. The sources are not well integrated, the report mostly reads as a disjointed collection of knowledge. Some inconsistencies and/or mistakes in the references list.	Draws on some relevant theoretical and practical tools related to ethics in data science (possibly including own experience) and there is at least one source not from the teaching material. Tools and resourced are used correctly, with only minor mistakes, and the report reads as a coherent piece of work. The references list is consistent.	Draws on the relevant theoretical and practical tools from the module as well as on some external sources. Tools and resources are used correctly, the external sources are integrated very well, and the report reads as a coherent piece of work. The references list is consistent.	Excellent work that not only uses the course materials but incorporates multiple external sources well. The knowledge and topic understanding demonstrated go substantially beyond the taught material. The references list is consistent.
Criterion C (default: 20%; range: 10% to 30%). Engages critically with	Statements in the source materials are taken at face value without further investigation.	Statements in the source materials are taken at face value without further investigation.	Minimal examination and critical engagement with the sources.	Multiple attempts are made to question and evaluate the quality of sources	Careful analysis and critical engagement with the sources. Appropriately discusses the	Exceptional work of a detailed and well-argued analysis that engages with the conflicting

	0-29% Fail	30-49% Fail	50-59% Good	60-69% Very Good	70-79% Excellent	80-100% Exceptional
the sources and arguments included (breadth and depth of analysis).	Little to no analysis of the topic.	Shallow or one-sided analysis of the topic.	Some mistakes and/or some sources accepted uncritically. Limited appreciation of or engagement with conflicting positions and perspectives.	and arguments used. A few argumentative mistakes when addressing the sources. The engagement with conflicting positions and perspectives is satisfactory but could be improved.	multiple conflicting positions and perspectives involved.	positions and perspectives involved. The critical analysis is at a publishable level.
Criterion D (default: 20%; range: 15% to 30%.) Discusses the topic in a way that is specific and connects the theory with concrete issues.	The report is overly generic and the theoretical elements are not connected at all to concrete issues. So, overall the report is (almost) entirely skewed more towards theoretical elements than concrete issues. No or very generic recommendations.	The report is mostly generic. There are attempts to connect the theory to concrete issues, but are mostly ineffective. That is, most connections are superficial, generic, tangential or convoluted. So, overall the theoretical elements	The report is mostly generic, but the attempts to connect the theory to concrete issues are sufficient and appropriate. Some connections are superficial, generic, tangential or convoluted. So, overall there is more of a skew towards theoretical elements than concrete issues.	There are multiple attempts to connect the theory to concrete issues and most of them are successful. Some connections may have some superficial, generic or convoluted elements. So, overall theoretical elements and concrete issues are sufficiently balanced.	Excellent links are made between theory and practice for the majority of the report, in a way that is mostly specific to the topic at hand. So, overall theoretical elements and concrete issues are very well balanced. Mostly specific and actionable recommendations.	Exceptional links are made between theory and practice for the entirety of the report, in a way that is fully specific to the topic at hand. So, overall theoretical elements and concrete issues are perfectly balanced. Fully specific and actionable recommendations.

	0-29% Fail	30-49% Fail	50-59% Good	60-69% Very Good	70-79% Excellent	80-100% Exceptional
		dominate over the concrete issues. No or very generic recommendations.	At least one specific and actionable recommendation.	Mostly specific and actionable recommendations.		
Criterion E (default: 15%; range: 10% to 20%.) The group discusses their work in a manner that is accurate, appropriate and clear.	No poster, or the poster is completely or largely unrelated to the assessment.	The poster is too busy, or too sparse, or mostly unclear or incomplete. It is a disjointed collection of perspectives. It shows little understanding of data ethics as a domain.	The content of the poster is sufficiently well selected and clear enough (in structure, style and content). Perspectives are compared, albeit minimally. Some elements, and/or the related understanding of data ethics as a domain, may be partially incomplete, superficial or with (major) mistakes.	The content of the poster is well selected and mostly clear (in structure, style and content). Comparison of perspective is thoughtful, but mostly factual. There is a thoughtful engagement with data ethics as a domain but may have limitations, or minor mistakes.	The content of the poster is very well selected and very clear (in structure, style and content). Comparison of perspective is thoughtful and adds some insights that would not be available via the individual perspectives only. There is a reasoned and specific engagement with data ethics as a domain, but there can be some, mostly minor, mistakes.	The content of the poster is perfectly selected and clear (in structure, style and content). Comparison of perspective is thoughtful and adds substantial insights that would not be available via the individual perspectives only. There is a fully reasoned and specific engagement with data ethics as a domain.
Criterion F (5%) Includes a suitable and	No self-reflection present. OR: self-reflection present, but out of	No self-reflection present. OR: self-reflection present, but out of	Self-reflection is included, but is generic and/or	Self-reflection is included, but is generic and/or	Self-reflection included is thoughtful, specific and complete.	Self-reflection included is thoughtful,

	0-29% Fail	30-49% Fail	50-59% Good	60-69% Very Good	70-79% Excellent	80-100% Exceptional
critical self-reflection.	scope or completely unsubstantiated.	scope or completely unsubstantiated.	mostly unsubstantiated.	partially unsubstantiated.		specific, complete and well argued.

On Moodle you will find an indicative template marking sheet. You can download it to see how the inner workings of the allocation of marks and weights across all criteria. However, the exact format in which the written summative feedback will be provided on Moodle may differ (that is, it may be provided as paragraphs in the comment box rather than as a file).